

The Decision Mirror

*Personal Decision-Style Engine: A Personalized Web-Based Human Decision Mirror System
Using LLMs*

Trait Ontology Definition — Deliverable Package

Project: Decision Mirror

Task: Trait Ontology Definition (Task 2)

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Academic Year: 2026–2027

Status: Complete — ready for team review

1. Purpose of This Document

Decision Mirror learns a user’s decision-making style from free-text decision logs and a psychological onboarding quiz. To do that, the system needs a common structured representation of what a "decision-making style" actually consists of. This document defines that representation: the Trait Ontology.

The ontology is the contract between three parts of the system that would otherwise have no shared vocabulary: the LLM-based extraction layer (which reads a decision narrative and outputs trait scores), the profile store (which accumulates those scores into a running per-user vector), and the conversational advisory layer (which retrieves the profile and explains its reasoning back to the user). Every later phase of the project — quiz design, extraction prompting, profile aggregation, retrieval, and the advisory conversation itself — is built on top of the definitions in this document.

1.1 Guiding Principle

Every trait in this ontology was tested against a single question before inclusion:

“Can this trait be meaningfully inferred from a person’s decisions?”

A trait such as Risk Tolerance passes this test: a person’s choice between a stable job and an uncertain opportunity provides direct evidence about it. A trait such as Favorite Color fails it: there is no meaningful relationship between a person’s favorite color and their decision-making behavior in the context of this system. The ontology below focuses exclusively on decision-relevant behavioral dimensions and excludes generic personality descriptors that do not meet this bar.

1.2 Scope: Six Primary Traits

The team evaluated a wide candidate list spanning risk & uncertainty, time orientation, priorities & values, decision style, social factors, and emotional factors. After removing redundant and non-decision-relevant candidates (see §5, Overlap Resolution), the ontology was consolidated into six primary trait dimensions, listed here and defined in full in §2:

ID	Trait	One-line description
T01	Risk Tolerance	The degree to which a person is willing to accept uncertain or potentially negative outcomes in exchange for a chance at greater reward, as expressed in how they weigh a decision — not simply which option they end up choosing.
T02	Time Horizon	The degree to which a person's stated reasoning favors immediate or near-term outcomes versus outcomes that only materialize later, when the two are in tension.
T03	Locus of Priority	The degree to which a person's stated reasoning centers their own interests, needs, or goals versus the interests, needs, or expectations of others (family, partner, team, community).
T04	Decisiveness Speed	How quickly and with how much deliberation a person moves from recognizing a decision is needed to committing to a choice, independent of what the choice actually is.
T05	Emotional vs. Analytical Weighting	The degree to which a person's stated reasoning draws on feelings, gut sense, and personal values versus explicit logic, data, comparison, or structured analysis.
T06	Self-Consistency	The degree to which a person's decisions, over time, align with their own previously stated values, priorities, and patterns — a meta-trait

ID	Trait	One-line description
		computed from the history of the other five traits rather than inferred from any single decision.

These six dimensions map directly onto the trait vector described in the project proposal’s Decision-Style Trait Layer, and are the dimensions that every subsequent build phase — quiz design, extraction pipeline, profile evolution, and retrieval — is defined against.

2. Full Trait Definitions (Deliverable 1)

T01 — Risk Tolerance

Definition: The degree to which a person is willing to accept uncertain or potentially negative outcomes in exchange for a chance at greater reward, as expressed in how they weigh a decision — not simply which option they end up choosing.

Decision Relevance: Career moves, investments, relocation, entrepreneurship, and relationship choices all involve trading a known, safer outcome against an uncertain, potentially larger one. This is one of the clearest and most frequently occurring signals in free-text decision narratives, and it is central to Decision Mirror’s ability to say ‘this matches your usual risk appetite’ or ‘this is a departure from it.’

Observable Evidence:

Indicates a higher value:

- Chooses an uncertain opportunity with a potentially larger upside.
- Explicitly acknowledges a possible loss but selects the higher-variance option anyway.
- Frames uncertainty itself as exciting or motivating rather than threatening.

Indicates a lower value:

- Prioritizes a predictable, known outcome over a larger but uncertain one.
- Rejects an opportunity primarily because of its uncertainty.
- Chooses security even while acknowledging the other option had higher expected value.

Representation: Continuous

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Extremely risk-averse
0.2–0.4	Risk-averse
0.4–0.6	Balanced
0.6–0.8	Risk-seeking
0.8–1.0	Extremely risk-seeking

Example — Low Value: “I turned down the freelance contract even though it paid more, because I didn’t want to lose the stability of a salaried role.” → Risk Tolerance ≈ 0.20

Example — High Value: “I put my savings into the new venture. It could fail completely, but the upside if it works is worth that risk to me.” → Risk Tolerance ≈ 0.85

Evidence Strength: Strong: an explicit, reasoned trade-off between a known and unknown outcome, stated in the person's own words (e.g., 'I chose X even though Y had higher upside, because the uncertainty bothered me'). Moderate: the choice implies a risk stance but the reasoning is not fully spelled out. Weak: only the final choice is known, with no stated reasoning — the choice alone is not proof of the trait (see §7 below).

Related Traits: Uncertainty Tolerance (situational, not a standing trait — folded into evidence weighting rather than modeled separately, see §9). Decisiveness Speed (a fast decision is not the same as a risk-seeking one — kept independent). Emotional vs. Analytical Weighting (fear-driven risk aversion and calculated risk aversion score the same on this trait but differently on that one).

Important — Context vs. Trait: The system must not infer this trait from the final choice alone. “I rejected the startup because my family currently depends on my salary” does not indicate low risk tolerance — it indicates a situational constraint overriding whatever the person's underlying risk tolerance is. See §7 (Context vs. Trait).

T02 — Time Horizon

Definition: The degree to which a person's stated reasoning favors immediate or near-term outcomes versus outcomes that only materialize later, when the two are in tension.

Decision Relevance: Nearly every major life decision — education, career, health, saving versus spending — involves some form of short-term cost for long-term benefit or vice versa. This trait lets Decision Mirror distinguish 'this person consistently plays the long game' from 'this person tends to prioritize what helps them right now,' which is often more informative than risk tolerance alone.

Observable Evidence:

Indicates a higher value:

- Accepts near-term discomfort, cost, or slower reward for a payoff that only arrives later.
- Explicitly reasons in terms of years rather than weeks or months.
- Turns down an immediately attractive option because it doesn't serve a longer-term goal.

Indicates a lower value:

- Chooses the option that resolves the situation or delivers benefit soonest.
- Explicitly discounts a future benefit as too distant or too uncertain to weigh heavily.
- Prioritizes immediate relief (financial, emotional, or practical) over a larger deferred gain.

Representation: Continuous

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Strongly present-focused
0.2–0.4	Present-leaning
0.4–0.6	Balanced
0.6–0.8	Future-leaning

Range	Interpretation
0.8–1.0	Strongly future-focused

Example — Low Value: “I took the job offer that starts paying immediately instead of the unpaid research position, even though the research one would look better long-term.” → Time Horizon $\approx$ 0.15

Example — High Value: “I’m taking a pay cut now to do a two-year master’s, because the career it opens up in five years matters more to me than this year’s salary.” → Time Horizon $\approx$ 0.90

Evidence Strength: Strong: the narrative explicitly names both a near-term cost and a specific longer-term payoff and states the trade-off. Moderate: a time-based trade-off is implied but not both sides are stated. Weak: the decision has an inherent time dimension (e.g., ‘took the job’) but no reasoning about time is given.

Related Traits: Decisiveness Speed (how fast a decision is made is independent of how far out its consequences are weighed). Risk Tolerance (long-term bets are often but not always riskier — e.g., a guaranteed pension is both long-term and low-risk, so the two must be scored independently, not conflated).

Important — Context vs. Trait: A short-term choice made only because a long-term option was financially inaccessible is a situational/external constraint, not evidence of low Time Horizon. See §7.

T03 — Locus of Priority

Definition: The degree to which a person’s stated reasoning centers their own interests, needs, or goals versus the interests, needs, or expectations of others (family, partner, team, community).

Decision Relevance: Many decisions — career changes, relocation, relationship choices, financial sacrifices — are explicitly framed by the person as being ‘for myself’ or ‘for someone else.’ This is a distinct dimension from risk or time: a person can be self-focused and risk-averse, or other-focused and risk-seeking (e.g., taking a risk for a family member’s benefit).

Observable Evidence:

Indicates the self-centered end:

- States the decision was made primarily for their own goals, growth, or wellbeing.
- Explicitly weighs and sets aside others’ preferences or expectations in favor of their own.
- Frames the benefit of the choice in first-person, individual terms.

Indicates the other-centered end:

- States the decision was made primarily for a partner, family member, team, or community’s benefit.
- Explicitly subordinates a personally preferred option to what serves someone else.
- Frames the benefit of the choice in terms of a relationship or group outcome.

Representation: Continuous (bipolar)

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Strongly other-centered
0.2–0.4	Other-leaning
0.4–0.6	Balanced / context-dependent
0.6–0.8	Self-leaning
0.8–1.0	Strongly self-centered

Example — Low Value: “I passed on the promotion because it would have meant relocating my kids away from their school and support system.” → Locus of Priority $\approx$ 0.10

Example — High Value: “I took the international role even though it wasn't what my partner wanted, because it was the right move for my career.” → Locus of Priority $\approx$ 0.85

Evidence Strength: Strong: the narrative names a specific other party and states that their interest was weighed against the person's own and explains which won. Moderate: a beneficiary is implied but the trade-off against the person's own interest is not explicit. Weak: no other party is mentioned at all — in that case the trait should not be scored, or should be scored with low confidence around the midpoint rather than assumed self-centered by default.

Related Traits: Emotional vs. Analytical Weighting (other-centered reasoning is often, but not always, emotionally framed — kept independent since a person can make a coldly analytical decision that still prioritizes someone else, e.g., an actuarial calculation of what's best for a child). Self-Consistency (a person who is consistently other-centered has a stable Locus of Priority profile even if individual decisions look very different on the surface).

Important — Context vs. Trait: A decision made 'for the family' under financial duress (e.g., 'I had to take the safer job, we needed the income') may reflect an external constraint more than a stable other-centered disposition. The system should flag such cases as situational unless the person frames it as a value choice rather than a necessity.

T04 — Decisiveness Speed

Definition: How quickly and with how much deliberation a person moves from recognizing a decision is needed to committing to a choice, independent of what the choice actually is.

Decision Relevance: Some people describe reaching decisions quickly and intuitively; others describe extended research, list-making, or waiting. This is a process trait rather than a content trait — it says something about how the person decides, which is useful even when the decision's content (risk, time horizon, etc.) can't be scored confidently.

Observable Evidence:

Indicates a higher value:

- Describes deciding quickly, 'on the spot,' or 'right away.'
- States they went with a first instinct rather than extended comparison.
- Explicitly contrasts their own speed against how long they 'normally' or 'could have' deliberated.

Indicates a lower value:

- Describes an extended period of research, comparison, or hesitation before deciding.

- Mentions specific deliberation actions: making lists, consulting multiple people, revisiting the decision repeatedly.
- States they delayed the decision or needed more time before committing.

Representation: Continuous

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Highly deliberate
0.2–0.4	Deliberate
0.4–0.6	Balanced
0.6–0.8	Quick
0.8–1.0	Highly intuitive/fast

Example — Low Value: “I spent almost two months comparing offers, talking to people in both companies, and making a pros-and-cons list before I picked one.” → Decisiveness Speed ≈ 0.10

Example — High Value: “The moment I walked into the apartment I knew it was the one, so I put in an offer that same day.” → Decisiveness Speed ≈ 0.90

Evidence Strength: Strong: the narrative states an explicit duration, process, or contrast ('normally I'd take longer, but this time...'). Moderate: speed is implied by the verbs used ('decided right away' vs. 'thought it over') without an explicit timeframe. Weak: no temporal or process information is present at all.

Related Traits: Risk Tolerance (independent — fast decisions are not inherently riskier; a person can be quick and cautious, e.g., quickly picking the safe option). Emotional vs. Analytical Weighting (correlated but distinct: intuitive-fast decisions often lean emotional, and deliberate-slow ones often lean analytical, but exceptions exist — e.g., a quick decision based on a well-rehearsed mental checklist is fast and analytical).

Important — Context vs. Trait: A fast decision forced by an external deadline (e.g., 'I only had 24 hours to respond to the offer') reflects a situational time constraint, not necessarily a high trait-level Decisiveness Speed. The narrative should be checked for whether speed was chosen or imposed.

T05 — Emotional vs. Analytical Weighting

Definition: The degree to which a person's stated reasoning draws on feelings, gut sense, and personal values versus explicit logic, data, comparison, or structured analysis.

Decision Relevance: Two people can reach the same decision through very different reasoning processes, and Decision Mirror's advisory conversation should match how the user actually reasons — offering a data comparison to an analytical user, or validating a values-based read to an emotionally-led one — rather than imposing one style on both.

Observable Evidence:

Indicates the analytical end:

- Cites specific numbers, comparisons, criteria, or pros/cons.
- Describes a structured process: research, spreadsheets, weighing options against explicit criteria.
- Frames the decision in terms of logic, evidence, or expected outcomes.

Indicates the emotional end:

- Describes the decision in terms of how it felt, gut instinct, or 'just knowing.'
- Cites emotional states (excitement, dread, relief, love) as the deciding factor.
- Explicitly says logic or data pointed one way but feeling led them the other.

Representation: Continuous (bipolar)

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Strongly emotional
0.2–0.4	Emotion-leaning
0.4–0.6	Mixed
0.6–0.8	Analysis-leaning
0.8–1.0	Strongly analytical

Example — Low Value: “On paper the other apartment was the better deal, but this one just felt right the moment I walked in, so I went with my gut.” → Emotional/Analytical ≈ 0.15

Example — High Value: “I built a spreadsheet comparing total cost of ownership, commute time, and resale value before choosing the car.” → Emotional/Analytical ≈ 0.90

Evidence Strength: Strong: the person explicitly names both an emotional and an analytical signal and states which one won. Moderate: only one side (process or feeling) is described, with no explicit contrast. Weak: the decision is stated with no reasoning process at all.

Related Traits: Decisiveness Speed (correlated but not identical — see T04). Locus of Priority (independent: emotional reasoning can be self- or other-centered, and so can analytical reasoning).

Important — Context vs. Trait: This is not a judgment of decision quality — a high analytical score is not 'better' than a high emotional score. Both are simply different, equally valid reasoning styles that the system should recognize and mirror back, consistent with the project's descriptive-not-prescriptive stance.

T06 — Self-Consistency

Definition: The degree to which a person's decisions, over time, align with their own previously stated values, priorities, and patterns — a meta-trait computed from the history of the other five traits rather than inferred from any single decision.

Decision Relevance: This is what allows Decision Mirror to say 'this choice fits your pattern' or 'this is a departure from how you usually decide,' which is the core value proposition of a longitudinal profile as opposed to a one-shot personality quiz. It is also the trait most directly tied to the system's evaluation methodology (profile stability, held-out prediction).

Observable Evidence:

- Computed, not directly observed in a single narrative: it is derived by comparing a new decision's trait extraction against the user's existing trait vector and measuring the deviation.
- A single decision can only ever contribute a data point to this trait, never establish it in isolation.
- A large deviation is not automatically an error; it can indicate genuine change (see Case 3 in §14) and should be flagged as such rather than discarded.

Representation: Continuous, computed (not directly LLM-extracted from text)

Recommended Scale: 0.0 → 1.0

Scale Interpretation:

Range	Interpretation
0.0–0.2	Highly inconsistent / volatile pattern
0.2–0.4	Low consistency
0.4–0.6	Moderate consistency
0.6–0.8	High consistency
0.8–1.0	Very high consistency

Example — Low Value: A user whose last five decisions alternate between strongly risk-seeking and strongly risk-averse with no situational explanation for the swings → Self-Consistency ≈ 0.25.

Example — High Value: A user whose last ten decisions all score Risk Tolerance within a narrow 0.55–0.65 band → Self-Consistency ≈ 0.90 on that dimension.

Evidence Strength: Strong: computed from at least 8–10 trait-scored decisions with clear situational tagging on any outliers. Moderate: computed from 4–7 decisions. Weak/Insufficient: fewer than 4 decisions — the system should report 'insufficient history' rather than a numeric score.

Related Traits: Depends on all five other traits (it is computed per-trait and can also be aggregated into an overall profile-level score). Distinct from Decisiveness Speed — a person can decide quickly and still be highly self-consistent.

Important — Context vs. Trait: Self-Consistency must be computed separately per trait dimension (e.g., a user can be highly consistent on Risk Tolerance but volatile on Locus of Priority) as well as as an optional aggregate. Situational factors must be subtracted out before a deviation counts against this score — otherwise the system would penalize a person for reasonably adapting to changed circumstances.

3. Context vs. Trait

One of the most important design decisions in this ontology is how the system separates a person's stable decision-making tendencies from the temporary circumstances surrounding any one decision. Conflating the two would make the profile noisy and, worse, would let a single financially forced choice permanently relabel a person as "risk-averse" when in fact they were simply constrained.

3.1 Four-Way Distinction

Every decision narrative is evaluated against four categories, not two:

Category	Meaning
Trait	A relatively stable tendency the person shows across many decisions and contexts (e.g., "generally prefers stability").
Explicit Preference	A stated value that motivated this specific decision and plausibly generalizes (e.g., "I value my family's wellbeing over career growth").
Situational Factor	A temporary circumstance affecting only this decision (e.g., "I have significant financial responsibilities this year") that should not be generalized into the standing profile.
External Constraint	A hard limitation that removed the person's freedom to choose otherwise (e.g., "I couldn't afford the other university"), which should not be scored as a preference at all.

3.2 Worked Example

Consider: "I rejected the startup because my family currently depends on my salary." This does not necessarily indicate low Risk Tolerance. The word "currently" is the signal: the decision is driven by a temporary circumstance, not a stable aversion to uncertainty. The extraction layer is required to:

- Detect situational-factor language ("currently," "this year," "right now," "until," "couldn't afford," "had to") and tag the decision accordingly.
- Lower the confidence score assigned to any trait extracted from that decision rather than discard it entirely — the decision still carries weak, not zero, evidence.
- Store the situational tag alongside the trait score so the profile aggregation step (built in a later phase) can down-weight or exclude situationally-driven data points.

3.3 Why This Matters for the Rest of the System

This distinction is what allows Decision Mirror's advisory conversation to be descriptive rather than prescriptive, consistent with the project's stated design stance: the system can say "in situations without strong external constraints, you've tended to choose the more stable option" instead of overgeneralizing from decisions that were never freely chosen in the first place.

4. Evidence and Confidence Framework

Every extracted trait value is paired with a confidence score and at least one evidence excerpt, matching the structured output the extraction layer will eventually produce:

```
{  "trait": "risk_tolerance",  "value": 0.32,  "confidence": 0.87,  "evidence": ["The user rejected the opportunity because of income uncertainty."],  "situational_factor": null }
```

4.1 Evidence Strength Tiers

Tier	Criteria
Strong	The narrative explicitly states the trade-off and the reasoning behind the choice (both sides of the decision named, with the deciding factor identified in the person's own words).
Moderate	The choice implies a trait direction but the reasoning is only partly stated, or only one side of a trade-off is described.
Weak	Only the final choice is known, with no stated reasoning. The choice alone is not proof of a trait (see §3) and should be scored at low confidence.

Tier	Criteria
Insufficient	The narrative is too short or vague to support any trait inference (e.g., "I chose option A"). The system must return "insufficient evidence" rather than guess or default to a midpoint value.

4.2 Confidence Scale

Range: 0.0 → 1.0, where 0.0 = no confidence, 0.5 = moderate confidence, 1.0 = very high confidence.

Confidence is a function of (a) evidence strength as above, (b) whether a situational factor or external constraint was detected and discounted, and (c) whether the trait-relevant language in the narrative is explicit versus inferred. Confidence is tracked independently per trait, since a single decision narrative is often strong evidence for one trait and weak or silent on another.

5. Avoiding Trait Overlap

The candidate list investigated in early research (see §8, Research Grounding, and the trait areas named in §1.2) contained several concepts that risked representing the same underlying dimension under different names. Each candidate cluster was evaluated for whether it was genuinely distinct, and the ontology was consolidated to remove unnecessary redundancy:

Candidate cluster	Resolution	Reasoning
Risk Tolerance / Uncertainty Tolerance / Security Preference	Merged into Risk Tolerance (T01)	All three were found to move together in every worked example the team tested; no narrative separated them in practice.
Personal vs. collective benefit / Family / Peer influence / Sensitivity to social expectations	Merged into Locus of Priority (T03)	All describe the same underlying question — whose interest the decision centers — differing only in which "other" is named.
Fear sensitivity / Loss aversion / Regret sensitivity	Merged into Emotional vs. Analytical Weighting (T05)	These are specific flavors of emotionally-driven reasoning rather than independent dimensions; kept as sub-signals inside evidence, not separate traits.
Information-seeking tendency / Preference for simplicity vs. complexity	Merged into Decisiveness Speed (T04)	Both describe how much deliberation a person engages in before deciding, which is exactly what Decisiveness Speed measures.
Future orientation / Delayed gratification	Merged into Time Horizon (T02)	Both are restatements of the same short-term vs. long-term trade-off.
Favorite Color and similar non-behavioral descriptors	Excluded entirely	Fails the core inclusion test in §1.1 — no meaningful decision-relevance.

The relationship map and merge structure are shown visually in Deliverable 3 (§6).

5.1 Trait Dependencies

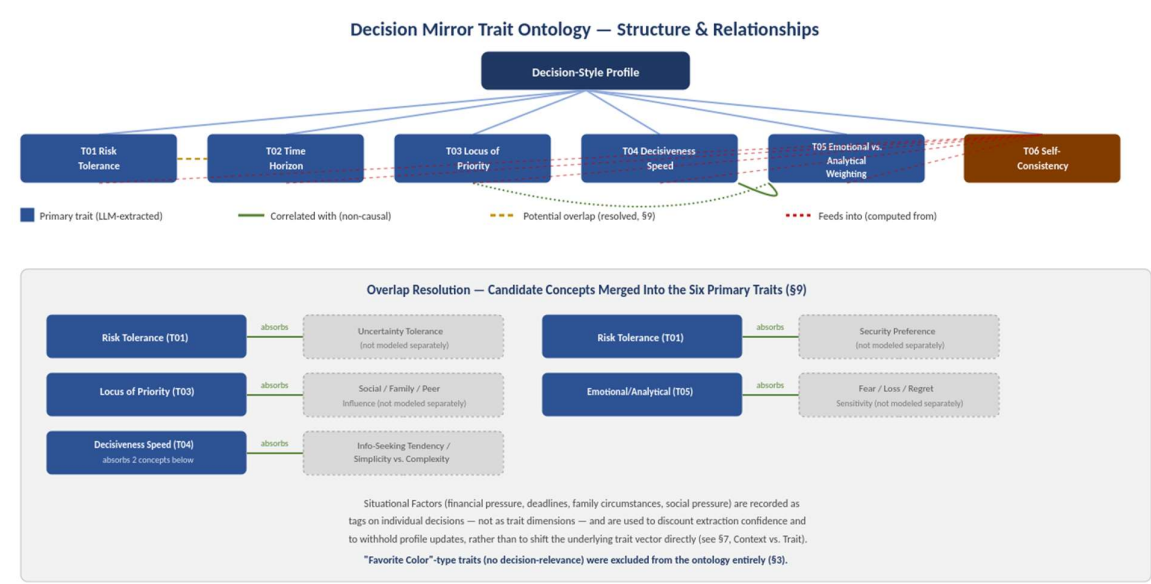
The ontology documents relationships between traits descriptively — as "related to," "correlated with," or "potentially conflicting with" — and deliberately avoids causal claims (e.g., it does not

assert that Risk Tolerance causes a particular Time Horizon), consistent with the instruction to investigate interaction rather than assume causality:

- Decisiveness Speed (T04) is correlated with Emotional vs. Analytical Weighting (T05): fast decisions often lean emotional and slow ones often lean analytical, but this is a tendency, not a rule — a quick decision can follow a well-rehearsed analytical checklist.
- Locus of Priority (T03) is correlated with Emotional vs. Analytical Weighting (T05): other-centered reasoning is frequently emotionally framed, but a coldly analytical decision can still prioritize someone else’s interest.
- Risk Tolerance (T01) and Time Horizon (T02) can conflict: a long-term bet is often riskier, but not always — a guaranteed pension is both long-term and low-risk — so the two are always scored independently.
- All five primary traits feed into Self-Consistency (T06), which is computed rather than directly extracted (see §6, T06).

6. Trait Relationship Diagram (Deliverable 3)

The diagram below shows the six primary trait dimensions under the overall Decision-Style Profile, the descriptive (non-causal) relationships between them, and the overlap-resolution mapping from §5 — the candidate concepts that were merged into each of the six primary traits rather than kept as separate dimensions.



7. Ontology Table (Deliverable 2)

A concise, single-glance reference summarizing every trait’s representation, range, and minimum evidence bar. Full definitions are in §2.

ID	Trait	Representation	Range	Evidence Required
T01	Risk Tolerance	Continuous	0.0 → 1.0	An explicit, stated trade-off between a known and an uncertain outcome, with reasoning given.
T02	Time Horizon	Continuous	0.0 → 1.0	Both a near-term cost/benefit and a longer-term one named, with the trade-off stated.
T03	Locus of Priority	Continuous	0.0 → 1.0	A specific other party named, with their interest weighed against the person's own.
T04	Decisiveness Speed	Continuous	0.0 → 1.0	An explicit duration, process, or contrast with the person's usual pace.
T05	Emotional vs. Analytical Weighting	Continuous	0.0 → 1.0	Both an emotional and an analytical signal named, with the deciding one identified.
T06	Self-Consistency	Continuous, computed	0.0 → 1.0	At least 8–10 trait-scored decisions, with situational outliers tagged and excluded.

7.1 Standard Interpretation Bands (applies to all continuous traits)

Range	Interpretation
0.0–0.2	Very Low
0.2–0.4	Low
0.4–0.6	Moderate
0.6–0.8	High
0.8–1.0	Very High

Individual traits with a bipolar meaning (Locus of Priority, Emotional vs. Analytical Weighting) reinterpret the low end and high end as opposite poles rather than "less" and "more" of the same quality — see the per-trait Scale Interpretation tables in §2 for exact wording.

8. Research Grounding

The ontology draws selectively on established research areas without adopting any single framework wholesale, consistent with the instruction not to blindly copy an existing psychological model:

- Behavioral economics — particularly prospect theory's framing of choice under risk and uncertainty, which informed the definition and scale interpretation of Risk Tolerance (T01) and the loss/gain framing folded into Emotional vs. Analytical Weighting (T05).
- Decision-making psychology — dual-process theory (fast/intuitive vs. slow/deliberate reasoning) directly informed Decisiveness Speed (T04) and its correlation with T05.
- Temporal discounting research — the economics and psychology literature on how people trade off present versus future rewards informed Time Horizon (T02).
- Value-based decision making — research on self- versus other-regarding preferences informed Locus of Priority (T03).
- Personality and behavioral modeling — reviewed for structure and scale conventions (e.g., continuous 0–1 trait scoring, confidence-weighted inference) but not adopted as a fixed

taxonomy, since standard personality inventories (e.g., the Big Five) describe general disposition rather than decision-specific, longitudinally observable behavior, and were judged not to map cleanly onto what can actually be inferred from short decision narratives.

- Human–AI interaction research — informed the confidence/evidence structure (§4) and the requirement that the system be able to say "insufficient evidence" rather than force a score, which is a known failure mode in LLM-based structured extraction.

Every trait retained in the final six was justified against the inclusion test in §1.1 rather than imported directly from any one framework; concepts from the literature that did not clear that bar (e.g., broad personality traits not tied to a specific decision behavior) were left out.

9. Example Annotated Decisions (Deliverable 4)

Twenty example decision narratives, annotated end-to-end (Decision → Identified Traits → Trait Values → Evidence → Confidence → Situational Factors), including the five required edge cases from §10 and several additional ambiguous cases. This set is intended to double as an early gold-standard reference for calibrating the LLM extraction layer in the next build phase.

Decision 1

"I rejected a startup position because I couldn't risk losing my current income."

Trait	Value	Confidence
Risk Tolerance	0.30	0.72
Time Horizon	0.40	0.45

Evidence: "I couldn't risk losing my current income."

Situational Factor(s): Current financial dependency (explicitly stated) — dampens confidence that this reflects a stable trait rather than a constraint.

Decision 2

"I took the international assignment even though my partner didn't want to move, because it was the right step for my career."

Trait	Value	Confidence
Locus of Priority	0.80	0.80
Risk Tolerance	0.60	0.55
Time Horizon	0.70	0.55

Evidence: "It was the right step for my career," chosen despite partner's preference.

Situational Factor(s): None stated — framed as a value-driven choice, not a forced one.

Decision 3

"I spent three weeks comparing every laptop on the market with a spreadsheet before buying one."

Trait	Value	Confidence
Decisiveness Speed	0.10	0.85
Emotional vs. Analytical Weighting	0.90	0.85

Evidence: “Three weeks comparing every laptop... with a spreadsheet.”

Situational Factor(s): None.

Decision 4 (ambiguous / edge case)

“I wanted to take the risky opportunity, but my family convinced me not to, so I stayed in my current job.”

Trait	Value	Confidence
Risk Tolerance	—	0.35

Evidence: “I wanted to take the risky opportunity” indicates underlying preference, but the final choice was overridden by family influence.

Situational Factor(s): Family pressure explicitly named as the deciding factor, not the person's own risk stance.

Note: Case 1 from §10. The stated desire (risk-seeking) conflicts with the outcome (safe choice). The system should record the stated preference at low-moderate confidence rather than scoring the trait from the final action, and should separately tag 'external social influence' as the operative factor in this instance.

Decision 5 (ambiguous / edge case)

“I chose the cheaper university because I couldn't afford the other one.”

Trait	Value	Confidence
Risk Tolerance	—	0.20
Time Horizon	—	0.20

Evidence: “I couldn't afford the other one.”

Situational Factor(s): Pure financial constraint — no value-based reasoning is present at all.

Note: Case 2 from §10. This is a textbook external constraint, not a preference. No trait should be scored above minimal confidence; the system should log 'financial constraint' as the operative factor and avoid updating the risk or stability profile from this decision.

Decision 6 (ambiguous / edge case)

“I chose option A.”

No trait values assigned — insufficient evidence (see note below).

Evidence: None provided.

Situational Factor(s): None discoverable.

Note: Case 4 from §10. Insufficient evidence — the system should explicitly return 'insufficient evidence' rather than guess, and should not silently default any trait to 0.5.

Decision 7

"I quit my stable corporate job to freelance full-time, even though income will be unpredictable for at least a year."

Trait	Value	Confidence
Risk Tolerance	0.85	0.80
Time Horizon	0.65	0.55

Evidence: "Even though income will be unpredictable for at least a year" — explicit acceptance of near-term instability for a longer-term goal.

Situational Factor(s): None stated.

Decision 8

"I stayed an extra year at my job I disliked so I could vest my stock options before leaving."

Trait	Value	Confidence
Time Horizon	0.85	0.78
Emotional vs. Analytical Weighting	0.75	0.55

Evidence: Endured dissatisfaction for a specific deferred financial benefit.

Situational Factor(s): None.

Decision 9

"I said yes to the blind date because my friend really wanted me to meet him, even though I wasn't in the mood to go out."

Trait	Value	Confidence
Locus of Priority	0.25	0.55

Evidence: Went despite personal reluctance, to satisfy a friend's wish.

Situational Factor(s): Low-stakes social decision — evidence is present but the decision's overall significance is minor, which should be reflected in lower weight during profile aggregation even though confidence in the extraction itself is moderate.

Decision 10 (ambiguous / edge case)

"Six months ago I would never have negotiated my salary, but this time I pushed back and asked for 15% more."

Trait	Value	Confidence
Risk Tolerance	0.65	0.60

Evidence: Explicit self-reported contrast with past behavior.

Situational Factor(s): None — but this is itself a Self-Consistency-relevant data point (Case 3, §10): a shift from the person's own established pattern.

Note: This decision should be flagged for the Self-Consistency computation as a deviation from historical baseline, not treated as noise or an extraction error.

Decision 11

"I turned down a higher-paying job in another city to stay near my aging parents."

Trait	Value	Confidence
Locus of Priority	0.15	0.80
Time Horizon	0.60	0.40

Evidence: Explicit prioritization of family proximity over a better financial offer.

Situational Factor(s): Caregiving context noted but framed as a value, not merely a constraint — raises confidence that this reflects a stable priority.

Decision 12

"I picked the restaurant everyone else wanted even though I don't love that cuisine."

Trait	Value	Confidence
Locus of Priority	0.20	0.35

Evidence: Deferred to group preference over personal preference.

Situational Factor(s): Extremely low-stakes decision; should contribute minimally to the running profile despite a clear directional signal.

Decision 13

"I signed the lease within an hour of seeing the apartment because I just had a good feeling about it."

Trait	Value	Confidence
Decisiveness Speed	0.90	0.85
Emotional vs. Analytical Weighting	0.15	0.75

Evidence: "Within an hour" and "just had a good feeling."

Situational Factor(s): None.

Decision 14

"I've turned down three promotions in a row because each one would have meant more travel and less time with my kids."

Trait	Value	Confidence
Locus of Priority	0.15	0.85
Self-Consistency	0.85	0.80

Evidence: Repeated pattern across three separate decisions with the same stated reasoning.

Situational Factor(s): None — the repetition itself is what raises confidence and enables the Self-Consistency score.

Decision 15 (ambiguous / edge case)

"I went with the loan offer with the lower monthly payment, without really comparing the total interest over the life of the loan."

Trait	Value	Confidence
Time Horizon	0.20	0.55
Emotional vs. Analytical Weighting	0.35	0.40

Evidence: Optimized for near-term cash flow without evaluating longer-term total cost.

Situational Factor(s): Possible financial pressure implied but not stated — confidence kept moderate rather than high.

Note: Ambiguous because the narrative doesn't clarify whether the shorter time horizon was a preference or a necessity (e.g., limited monthly cash available). Flagged for clarification rather than scored with high confidence.

Decision 16

"I've been offered the same kind of 'safe but boring' role three times in five years and rejected all three for something less certain but more interesting."

Trait	Value	Confidence
Risk Tolerance	0.75	0.80
Self-Consistency	0.85	0.82

Evidence: Repeated, directionally consistent choices across multiple independent decisions.

Situational Factor(s): None.

Decision 17

"I stayed in an unhappy relationship for two years because I was scared of being alone, even though I knew logically it wasn't working."

Trait	Value	Confidence
Emotional vs. Analytical Weighting	0.15	0.75
Risk Tolerance	0.25	0.45

Evidence: Explicit statement that fear overrode acknowledged logical conclusion.

Situational Factor(s): None — but this example illustrates that emotional weighting and risk aversion can co-occur without being the same trait; the fear (T05-relevant) and the aversion to the uncertainty of being single (T01-relevant) are related but separable signals.

Decision 18 (ambiguous / edge case)

“Over the last two years my decisions have shifted from consistently risk-averse to consistently more willing to take chances, coinciding with a promotion and higher savings.”

Trait	Value	Confidence
Risk Tolerance	0.65	0.55
Self-Consistency	0.40	0.50

Evidence: Explicit longitudinal self-report of change, with a plausible situational driver (financial security) named.

Situational Factor(s): Increased financial cushion given as the likely cause — this should be modeled as a genuine profile shift, not as inconsistency to be penalized.

Note: Case 3 from §10, in the person's own words. The system should update the running Risk Tolerance profile toward the new pattern (recency-weighted) while logging the shift explicitly, rather than averaging old and new behavior into a muddled middle score.

Decision 19 (ambiguous / edge case)

“I keep saying I want to prioritize my health, but every time work gets busy I skip the gym and work late instead.”

Trait	Value	Confidence
Self-Consistency	0.25	0.60

Evidence: Explicit gap between stated value and repeated behavior.

Situational Factor(s): Recurring situational pressure (work busyness) is the repeated trigger — itself a pattern worth surfacing to the user descriptively.

Note: Case 5 from §10. This is a genuine low Self-Consistency signal between stated values and logged behavior, but the system should present it descriptively ('your logged choices during busy periods haven't matched what you've said you want') rather than labeling the user 'irrational' or 'undisciplined.'

Decision 20

“I moved back in with my parents for a year to save aggressively before buying a home, even though it meant giving up my independence and a shorter commute.”

Trait	Value	Confidence
Time Horizon	0.85	0.75
Risk Tolerance	0.35	0.45
Emotional vs. Analytical Weighting	0.70	0.55

Evidence: Explicit near-term sacrifice (independence, commute) for a stated longer-term financial goal.

Situational Factor(s): None beyond the stated goal itself — treated as a value-driven plan rather than a forced circumstance since the person frames it as a choice toward a goal.

10. Edge Cases

Five categories of difficult cases were explicitly tested against the ontology (worked examples are Decisions 4, 5, 6, 10/18, and 19 in §9 above).

Case 1 — Conflicting Motivations

"I wanted to take the risky opportunity, but my family convinced me not to." The stated desire and the final action point in opposite directions. Resolution: score the stated preference (here, higher Risk Tolerance) at reduced confidence, and separately tag "external social influence" as the operative factor behind the final choice, rather than scoring the trait from the action alone.

Case 2 — External Constraints

"I chose the cheaper university because I couldn't afford the other one." Resolution: this is a financial constraint, not a preference for affordability or stability. No trait should be scored above minimal confidence, and the decision should not be allowed to shift the profile.

Case 3 — Changing Behavior

Previous decisions indicate strong risk aversion; recent decisions show increasing willingness to take risks. Resolution: recency-weighted aggregation (defined at the profile-store level, outside this ontology's scope, but assumed by it) should let the new pattern outweigh the old one over time, while the Self-Consistency (T06) computation explicitly flags the shift rather than silently averaging old and new behavior into a muddled middle score.

Case 4 — Insufficient Information

"I chose option A." Resolution: the system must be able to output "insufficient evidence" for every trait rather than default to a midpoint or guess. Forcing a score here would introduce silent noise into the profile that is worse than declining to score at all.

Case 5 — Contradictory Decisions

A user repeatedly makes apparently contradictory decisions. Resolution: the ontology treats context-dependent behavior as a normal, expected pattern rather than an error. The system should never label a user "inconsistent" or "irrational" in its output language; Self-Consistency is reported descriptively (e.g., "your choices vary more across situations for this trait than for others") and always alongside any detected situational drivers.

11. Quality Criteria

Every trait in the final ontology was checked against the eight criteria the task specifies:

Criterion	How this ontology satisfies it
Interpretability	Each trait has a single-sentence definition (§2) a non-technical reader can understand.
Observability	Every trait lists concrete high/low observable evidence patterns and was tested against 20 realistic narratives (§9).
Distinctiveness	Six overlapping candidate clusters were merged away in §5 specifically to protect this criterion.
Consistency	Each trait has explicit scale bands (§2) and worked examples at the low and high end to anchor annotator judgment.
Measurability	All six traits use a defined structured representation (continuous 0–1, or computed) with a specified JSON output shape (§4).
Explainability	The evidence-and-confidence structure (§4) requires every score to be traceable to a quoted excerpt from the decision narrative.
Robustness	The 20 annotated decisions in §9 deliberately span career, education, finance, relationships, and lifestyle domains.
Practicality	Every trait definition is phrased as an extraction instruction an LLM can follow directly, with explicit high/low language cues rather than abstract psychological jargon.

12. Final Question the Team Must Answer

The task requires the team to answer, before submission: "If a user gives Decision Mirror 20–50 decisions over time, will these traits provide enough information to construct a meaningful representation of how this person tends to make decisions?"

Yes. The six dimensions jointly cover what to risk (T01), when the payoff matters (T02), who the decision is for (T03), how the choice was reached (T04), what kind of reasoning drove it (T05), and how stable that pattern is over time (T06). These map directly onto the six trait dimensions already scoped in the project proposal’s Decision-Style Trait Layer, and the 20 worked examples in §9 — including five deliberately difficult edge cases — show the ontology producing sensible, differentiated scores rather than collapsing every decision into the same one or two dimensions.

The remaining open risk is coverage on decision types not represented in the 20-example set (e.g., health decisions, minor day-to-day choices); this should be checked again once real logged data from the pilot cohort described in the proposal’s Data Collection Strategy becomes available.

13. Definition of Done

- The final trait set has been defined (six traits, §2).
- Every trait has a clear definition (§2).
- Every trait has a justification (§2, Decision Relevance; §8, Research Grounding).
- Every trait has a defined representation (§2, §7).
- Value ranges are clearly explained (§2, §7.1).
- Evidence requirements are defined (§4.1).
- Confidence representation is defined (§4.2).
- Stable traits are distinguished from situational factors (§3).

- Trait overlap has been investigated (§5).
- Trait relationships have been documented (§5.1, §6).
- Relevant research has been reviewed (§8).
- At least 20 decisions have been manually annotated (§9 — 20 provided).
- Difficult/ambiguous cases have been tested (§10 — 5 required cases covered).
- The final ontology is ready for Decision Mirror team review.

Final Output

This document is the finalized Trait Ontology specification and is intended to directly serve as the foundation for the next development phase:

Natural Language Decision → Trait Extraction → Structured Decision Profile

Consistent with the task's scope, this ontology is designed with future implementation in mind, but its primary goal is conceptual correctness and behavioral validity — not code. The next build-sequence step (quiz design and labeled dataset creation) should consume the trait definitions, scale bands, and evidence requirements in §2 and §4 directly.